

Undergraduate in the Xu Teli Honors Program at Beijing Institute of Technology, majoring in Computer Science and Technology. My research interests center on **reinforcement learning with long-term memory, self-evolving models, recursive learning and self-improvement, multimodal foundation models, and model interpretability**. I am particularly interested in agents that learn from persistent memory and improve their own capabilities through recursive feedback loops.

Education

Beijing Institute of Technology

Sep 2024 – Present

[B.Eng. Candidate in Computer Science and Technology · Xu Teli Honors Program](#)

- Class representative of the Xu Teli Honors Program.
- CET-4 score: 572; Second Prize in the National English Competition for College Students.

Research Experience

Threader · Long-term Memory for Conversational Agents

Dec 2025 – May 2026

[Research collaboration with Tsinghua University · Co-author · NeurIPS 2026 under review](#)

- Co-developed Threader, a lossless long-term memory framework addressing information loss, latency, and cost in compression-based memory systems.
- Introduced a topic segmentation, multi-view encoding, and multi-signal retrieval pipeline that preserves complete conversations while improving construction and retrieval efficiency and reducing token cost.
- Outperformed representative systems including MemGPT and Mem0 on long-conversation memory benchmarks, improving long-range consistency and response efficiency.

SAIL · Brain-Model Semantic Alignment

Sep 2025 – May 2026

[Beijing Institute of Technology · Co-author · NeurIPS 2026 under review](#)

- Co-developed a sparse autoencoder (SAE) framework for aligning brain activity with interpretable representations inside multimodal foundation models.
- Disentangled high-dimensional visual features into monosemantic concepts and matched them to fMRI signals from the human visual cortex, revealing hierarchical correspondences with the ventral visual pathway.
- Demonstrated that SAE features outperform raw model features in brain-alignment tasks.

Industry Experience

ByteDance · Volcano Ark

Apr 2026 – Jun 2026

[Product Solutions Intern · AI Products & MaaS Solutions](#)

- Contributed to enterprise AI product and MaaS solution design by mapping foundation-model capabilities to business scenarios and product delivery paths.
- Supported solution research and product communication for B2B AI use cases, translating technical capabilities into actionable business proposals.

Selected Projects

Collector+ (Collector-AI)

Jan 2026

[FastAPI · React · LLM · Prompt Engineering](#)

- Built a low-latency information extraction and gamified reading assistant that transforms long-form articles and videos into binary predictive questions and supports source-grounded Q&A over saved content.
- Designed a linear monolithic asynchronous pipeline with approximately 15–30 seconds end-to-end generation latency and universal parsing for WeChat articles and Bilibili videos.
- [GitHub](#) · [Live Demo](#)

FitSnap

Next.js 16 · React 19 · DeepSeek-V3 · Zustand

- Transformed unstructured instructional videos into structured, timestamped steps with action categories and key parameters, with bidirectional synchronization between video playback and step cards.
- Built a content-agnostic generative UI that dynamically renders domain-specific components for fitness, cooking, and other instructional content.
- [Live Demo](#)

BiGEM Data Retrieval and Visualization Platform

Sep 2024 – Oct 2025

React · D3.js · Three.js · Python

- Cleaned and connected iGEM data from 2004–2024 and visualized global collaboration networks with force-directed graphs, trees, and Sankey diagrams.
- Integrated LLM-based semantic analysis to improve retrieval of biological parts and historical projects; received a 2025 iGEM Gold Medal and Best Software in the undergraduate category.
- [Source](#) · [Project Wiki](#)

Multi-Agent Shopping Assistant

Nov 2025 – Present

Python · Vue 3 · MCP · RAG

- Built an e-commerce assistant around the Model Context Protocol, implementing the LLM → Router → MCP Server → Backend API tool-calling pipeline.
- Developed a Vue 3 conversational interface that renders tool results as visual product cards.
- [GitHub](#)

Honors & Awards

iGEM Competition · Gold Medal and Best Software, Undergraduate Category	2025
China Undergraduate Mathematical Contest in Modeling · Beijing First Prize	2025
Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling · Honorable Mention	2025
National English Competition for College Students · Second Prize	2025
Beijing Institute of Technology · First- and Second-Class Scholarships	2025

Technical Skills

AI Engineering	PyTorch, Prompt Engineering, Structured Output, Function Calling, RAG, Long-context Management
Frontend & Visualization	Next.js, React, Vue 3, TypeScript, D3.js, Three.js, Framer Motion
Backend & Engineering	Python, FastAPI, Git, Docker, Linux, Data Pipelines
Languages	Chinese (Native), English (CET-4 572; Professional working proficiency)